

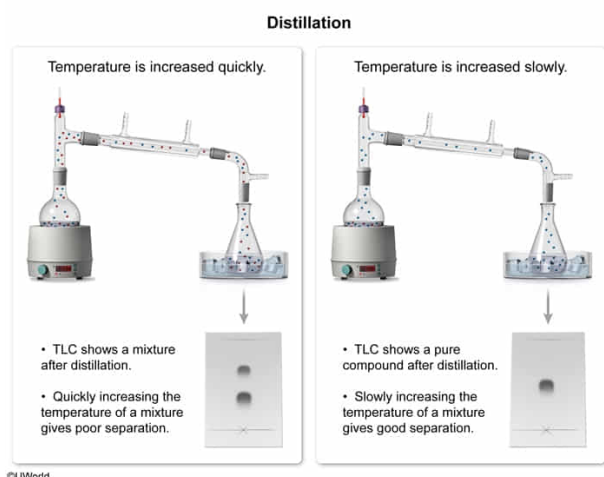
If thin-layer chromatography shows that a distillation of molecules with boiling points above 150 °C results in an impure compound, one can conclude that a possible error made during the distillation was:

- ☐ A. reducing the pressure of the distillation system. [17%]
- ☐ B. using a tall fractionating column. [6%]
- ✓ ☒ C. heating the mixture too quickly. [71%]
- ☐ D. placing boiling chips in the mixture. [4%]

Correct

71% answered correctly

Explanation:



Distillation is a technique used to **separate** liquid molecules based on their **boiling points**, and there are three common types: **simple**, **fractional**, and **vacuum**. In each, a liquid mixture is heated, and the molecule with the lowest boiling point distills first. The mixture must be **heated slowly** so that the molecule with the lowest boiling point can evaporate before the molecules with the higher boiling points. The vapors of the lower boiling-point compound travel up the distillation column (which is fitted with a thermometer), condense and travel through the condenser, and then drip into the receiving flask. The flask is placed in an ice bath to prevent the distillate from evaporating.

If thin-layer chromatography shows that the distillate is not pure, the distillation was not successful at separating the compounds in a mixture. Because a mixture should be heated at a slow rate for separation, a possible error in the distillation is **heating the mixture too quickly**, causing the molecules to evaporate together.

(Choice A) Compounds with boiling points above 150 °C *must* be distilled under reduced pressure using vacuum distillation. This decreases the boiling points, preventing decomposition of the compounds.

(Choice B) Fractional distillation uses a column to improve separation of compounds by increasing the length of the path the vapor must travel before condensing and dripping into the receiving flask.

(Choice D) Boiling chips are used to evenly heat a liquid and prevent [superheating](#). Their use in the distillation would prevent the mixture from abruptly forming large bubbles that could spill over into the receiving flask.

Educational objective:

Distillation is a technique that separates molecules based on their boiling points, and there are three main types: simple, fractional, and vacuum. The molecule with the lowest boiling point distills before the molecules with higher boiling points. The mixture must be heated slowly to ensure that a lower boiling-point compound evaporates before a higher boiling-point compound and that the molecules are separated.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Separation Techniques, Spectroscopy, and Analytical
Methods